

Cyclic Fibonacci sequences

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ABSTRACT

A cyclic Fibonacci sequence modulo n is a Fibonacci sequence $f_0, f_1, \dots$ satisfying the recursion $f_{i+1} = f_{i-1} + f_i$ such that $f_i \equiv f_j \pmod{n} \Rightarrow f_{i+1} \equiv f_{j+1} \pmod{n}$ for all $i, j \geq 0$. We prove that if $p \equiv \pm 1 \pmod{5}$ for every prime factor p of n , then a cyclic Fibonacci sequence modulo n exists. In addition, we study cyclic Fibonacci sequences in finite groups. In particular, we prove that, among non-cyclic abelian finite groups, only $\mathbb{Z}_2 \times \mathbb{Z}_2$ and $\mathbb{Z}_3 \times \mathbb{Z}_3$ admit a Fibonacci sequence that precisely includes the non-identity elements.

KEYWORDS

Cyclic Fibonacci sequence; coprime Fibonacci sequence; reduced Fibonacci sequence

1. Introduction

Given integers f_0, f_1 , define the associated *Fibonacci* sequence $f_0, f_1, \dots$ by the recurrence

$$f_{i+1} = f_{i-1} + f_i, \quad i \geq 1.$$

Let F_i , $i \geq 0$, denote the standard Fibonacci sequence, where $F_0 = 0$, $F_1 = 1$, and $F_{i+1} = F_i + F_{i-1}$, $i \geq 1$. Then every Fibonacci sequence satisfies

$$f_i = F_i f_1 + F_{i-1} f_0, \quad i \geq 0. \tag{1.1}$$

For a positive integer n , the least period of the standard Fibonacci sequence is called the *Pisano* period and is denoted by $\pi(n)$. Equation (1.1) implies that any Fibonacci sequence f_i , $i \geq 0$, is purely periodic; that is, there exists an integer k such that $f_{i+k} = f_i$ for all $i \geq 0$. Moreover, the least period of any Fibonacci sequence modulo n divides $\pi(n)$.

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Definition 1.1. Let $f_0, f_1, \dots$ be a Fibonacci sequence of integers and n be a positive integer.

- i) The sequence is called ‘cyclic’ (CF-sequence) modulo n if there are no repeated terms in a full period, or equivalently, for all $i, j \geq 0$:

$$f_i \equiv f_j \pmod{n} \implies f_{i+1} \equiv f_{j+1} \pmod{n}.$$

The length of a CF-sequence is the least positive k such that $f_k \equiv f_0 \pmod{n}$.

- ii) The sequence is called ‘coprime’ modulo n if $\gcd(f_i, n) = 1$ for all $i \geq 0$.
iii) The sequence is called ‘complete’ modulo n if its terms contain every congruence class modulo n .
iv) The sequence is called ‘reduced’ modulo n if its terms comprise precisely the nonzero congruence classes modulo n .

Example 1.2. The Fibonacci sequence $1, 8, \dots$ modulo 11 has period

$$1, 8, 9, 6, 4, 10, 3, 2, 5, 7,$$

which contains every nonzero residue modulo 11 exactly once. Hence it is a reduced and coprime cyclic Fibonacci sequence. By contrast, the Fibonacci sequence $1, 6, 7, \dots$ modulo 29 has period

$$1, 6, 7, 13, 20, 4, 24, 28, 23, 22, 16, 9, 25, 5,$$

whose terms are all distinct and relatively prime to 29 but do not exhaust the nonzero residue classes. Thus it is coprime and cyclic but not reduced.

Brison [2] determined all prime moduli for which a reduced CF-sequence exists. He used the term *complete Φ -sequence* and studied such sequences in the context of finite fields.

Theorem 1.3 (Brison [2] (1992).). *There exists a reduced CF-sequence modulo a prime p if and only if $p = 5$ or $\pi(p) = p - 1$.*

The Fibonacci roots modulo an integer n are the roots of $x^2 - x - 1$ modulo n . If x is a Fibonacci root, then the geometric sequence $1, x, x^2, \dots$ is a coprime CF-sequence modulo n . Consequently, if every prime divisor p of an integer n satisfies

$$p \equiv \pm 1 \pmod{5},$$

then coprime CF-sequences exist modulo n (Theorem 2.9). The converse, however, is false. For example, the Fibonacci sequence $1, 7, 8, \dots$ is a coprime CF-sequence modulo $377 = 13 \times 29$, even though $13 \not\equiv \pm 1 \pmod{5}$.

The discussion thus far extends naturally to the setting of finite groups. A sequence of elements $g_0, g_1, g_2, \dots$ in a group G is called a Fibonacci sequence if

$$g_{n+1} = g_{n-1}g_n,$$

for all $n \geq 0$. Knox [9] observed that every Fibonacci sequence in a finite group is purely periodic; i.e., there exists a positive integer k such that $g_{i+k} = g_i$ for all $i \geq 0$.

Definition 1.4. Let $g_0, g_1, \dots$ be a Fibonacci sequence in G .

- i) The sequence is called ‘cyclic’ (or CF-sequence) if there are no repeated terms in a full period, or equivalently, for all $i, j \geq 0$:

$$g_i = g_j \implies g_{i+1} = g_{j+1}.$$

We refer to the least period of a CF-sequence as its length.

- ii) The sequence is called ‘coprime’ if it is not included entirely in a proper subgroup of G .
iii) The sequence is called ‘complete’ if it contains every element of the group.
iv) The sequence is called ‘reduced’ if it contains precisely the non-identity elements.

Knox [9] showed that a Fibonacci sequence containing the identity is included in a cyclic subgroup. This together with Burr’s result [4] on complete Fibonacci sequences modulo integers implies the following theorem.

Theorem 1.5 (Burr-Knox [4, 9]). *The only finite groups that admit a complete Fibonacci sequence are the cyclic groups of order $n = 5^k m$, where $k \geq 0$ and*

$$m \in \{1, 2, 4, 6, 7, 14\} \cup \{3^j : j \geq 0\}.$$

Brison [2] proved that if $p \geq 7$, the additive group $(\mathbb{Z}_p)^k$ admits a reduced CF-sequence if and only if $k = 1$ and $\pi(p) = p - 1$. In Theorem 5.4, we show that non-cyclic abelian groups of size ≥ 10 do not admit a reduced Fibonacci sequence.

While every non-cyclic abelian group admits a CF-sequence (Theorem 5.2), we give infinitely many examples of non-cyclic abelian groups that do not admit coprime CF-sequences (Example 5.9).

Building on these results, our main contributions in this article can be summarized as follows.

Summary of results

- Every coprime CF-sequence modulo a prime p with length $\pi(p)$ is a coprime CF-sequence modulo p^k with length $\pi(p^k)$ for all $k \geq 1$ (Theorem 2.6).
- If every prime factor p of an integer n satisfies $p \equiv \pm 1 \pmod{5}$, then there exist at least 2^m coprime CF-sequences modulo n , where m is the number of prime factors of n (Theorem 2.9).
- The natural density of integers modulo which a coprime CF-sequence exists is zero (Theorem 3.3).
- Modulo an integer n there are at most two possible lengths for coprime Fibonacci sequences (Theorem 4.5).
- Every finite non-cyclic abelian group admits a cyclic Fibonacci sequence (Theorem 5.2).
- The only finite non-cyclic abelian groups that admit a reduced Fibonacci sequence are $\mathbb{Z}_2 \times \mathbb{Z}_2$ and $\mathbb{Z}_3 \times \mathbb{Z}_3$ (Theorem 5.4).

Section 2 is dedicated to CF-sequences modulo integers. In Section 3, we show that the natural density of integers modulo which a coprime CF-sequence exists is zero (Theorem 3.3). In Section 4, we study the period lengths of Fibonacci sequences modulo integers. In Section 5, we discuss Fibonacci sequences in finite groups in more detail

especially in the abelian case. Finally, in Section 6, we propose several open problems on CF-sequences.

2. Coprime CF-sequences Modulo Integers

For what values of n does there exist at least one coprime CF-sequence modulo n and how many *non-equivalent* coprime CF-sequences exist?

Definition 2.1. *Two integer sequences $f_0, f_1, \dots$ and $g_0, g_1, \dots$ are called equivalent modulo n if each sequence is an integer multiple of the other sequence modulo n ; more precisely, if there exist integers x, y such that $f_i \equiv xg_i \pmod{n}$ and $g_i \equiv yf_i \pmod{n}$ for all $i \geq 0$.*

We denote the number of non-equivalent coprime CF-sequences modulo n by $\mu(n)$.

Let $\text{ord}_n(a)$ denote the multiplicative order of a modulo n , that is, the least positive integer k such that $a^k \equiv 1 \pmod{n}$. The following lemma describes the relationship between the multiplicative orders of Fibonacci roots and the Pisano period.

Lemma 2.2. *Let p be a prime number such that $p \equiv \pm 1 \pmod{5}$, and let α, β be the roots of $x^2 - x - 1$ modulo p . Then $\text{ord}_p(\alpha)$ and $\text{ord}_p(\beta)$ are not both odd. Let α be such that $\text{ord}_p(\alpha)$ is even. Then for each $k \geq 1$ there exist $\alpha_k, \beta_k \in \mathbb{Z}_{p^k}$ such that α_k, β_k satisfy the equation $x^2 - x - 1 \equiv 0 \pmod{p^k}$, $\alpha_1 = \alpha$, $\beta_1 = \beta$, and*

- i) $\alpha_{k+1} \equiv \alpha_k \pmod{p^k}$ and $\beta_{k+1} \equiv \beta_k \pmod{p^k}$.*
- ii) If $\text{ord}_p(\beta)$ is odd, then $\pi(p^k) = \text{ord}_{p^k}(\alpha_k) = 2 \text{ord}_p(\beta_k) \equiv 2 \pmod{4}$.*
- iii) If $\text{ord}_p(\beta)$ is also even, then $\pi(p^k) = \text{ord}_{p^k}(\alpha_k) = \text{ord}_{p^k}(\beta_k) \equiv 0 \pmod{4}$.*

Proof. The case $k = 1$ follows from Theorem 2.5 of [2], and the general case follows from Lemmas 2.4 and 2.5 of [13]. $\square$

If $f_0, f_1, \dots$ is a Fibonacci sequence with the least period d modulo n , then $d \mid \pi(n)$. Hence the largest possible length of a CF-sequence modulo n is $\pi(n)$.

Definition 2.3. *A maximal CF-sequence $f_0, f_1, \dots$ modulo n is a CF-sequence that has length $\pi(n)$. We denote the number of non-equivalent maximal coprime CF-sequences modulo n by $\mu_0(n)$.*

We denote the p -adic valuation of integer n by $\nu_p(n)$, that is the largest integer k such that $p^k \mid n$. Also, let $z(p)$ denote the rank of apparition modulo p , that is the least positive integer such that $p \mid F_{z(p)}$. If $z(p) \mid i$ and $p \neq 2, 5$, then [11]

$$\nu_p(F_i) = \nu_p(i) + \nu_p(F_{z(p)}). \quad (2.1)$$

Lemma 2.4. *Let $p \neq 2, 5$ be a prime and $t = t(p)$ be the largest integer such that $\pi(p^t) = \pi(p)$. Then*

$$\pi(p^i) = \begin{cases} \pi(p) & \text{if } 1 \leq i \leq t(p); \\ p^{i-t(p)}\pi(p) & \text{if } i > t(p), \end{cases} \quad (2.2)$$

and

$$z(p^i) = \begin{cases} z(p) & \text{if } 1 \leq i \leq t(p); \\ p^{i-t(p)}z(p) & \text{if } i > t(p). \end{cases} \quad (2.3)$$

In addition, $t(p) = \nu_p(F_{z(p)})$.

Proof. Equations (2.2) and (2.3) follow from [15, 18]. To prove $t = \nu_p(F_{z(p)})$, we note that since $z(p^{t+1}) > z(p^t)$, we have $t \leq \nu_p(F_{z(p^t)}) < t+1$, and so $t = \nu_p(z_{\pi(p^t)}) = \nu_p(F_{z(p)})$. $\square$

The next lemma shows how increasing the Fibonacci index by a multiple of $\pi(n)$ modifies the term modulo $n = p^k$ for odd prime p .

Lemma 2.5. *Let p be an odd prime and $n = p^{k-1}$, $k > t$, where $t = t(p) = \nu_p(F_{z(p)})$ is the largest integer with $\pi(p^t) = \pi(p)$. Then for all $r, j \geq 0$:*

$$F_{r\pi(n)+j} \equiv F_j + arp^{k-1}L_j \pmod{p^k}, \quad (2.4)$$

where $a \not\equiv 0 \pmod{p}$ is a constant depending only on p , and $L_i, i \geq 0$, is the standard Lucas sequence defined by $L_0 = 2$, $L_1 = 1$, and $L_{i+1} = L_{i-1} + L_i$, $i \geq 0$.

Proof. Since $F_{\pi(n)} \equiv 0 \pmod{p^{k-1}}$, there exists an integer a such that $F_{\pi(n)} \equiv 2ap^{k-1} \pmod{p^k}$. Since $k > t$, by Lemma 2.4, we have $\pi(p^{k-1}) = p^{k-1-t}\pi(p)$. If $p \neq 5$, then

$$\nu_p(F_{\pi(n)}) = \nu_p(\pi(n)) + \nu_p(F_{z(p)}) = k-1-t + \nu_p(F_{z(p)}) = k-1.$$

If $p = 5$, then $\pi(n) = \pi(5^{k-1}) = 4 \times 5^{k-1}$, hence [11, Lemma 1]

$$\nu_p(F_{\pi(n)}) = \nu_p(\pi(n)) = k-1.$$

Therefore, for all odd primes p , we have $p \nmid a$.

Next, we deal with the base cases $j = 0, 1$.

First, suppose that $j = 0$. We prove the claim in this case by induction on $r \geq 0$. The base case follows from our choice of a . For the induction step, we note that

$$\begin{aligned} F_{(r+1)\pi(n)} &\equiv F_{\pi(n)+1}F_{r\pi(n)} + F_{\pi(n)}F_{r\pi(n)-1} \\ &\equiv F_{r\pi(n)} + F_{\pi(n)} \\ &\equiv 2arp^{k-1} + 2ap^{k-1} \\ &\equiv 2a(r+1)p^{k-1} \pmod{p^k}. \end{aligned}$$

Next, suppose that $j = 1$. We prove the claim in this case by induction on $r \geq 0$. For $r = 0$, we need to show that $F_{\pi(n)+1} \equiv 1 + ap^{k-1} \pmod{p^k}$. It follows from the equations

$$F_{\pi(n)+1}(2 - F_{\pi(n)+1}) = 1 - (F_{\pi(n)+1} - 1)^2 \equiv 1 \pmod{p^k}, \quad (2.5)$$

$$F_{\pi(n)+1}F_{\pi(n)-1} = F_{\pi(n)}^2 + (-1)^{\pi(n)} \equiv 1 \pmod{p^k}, \quad (2.6)$$

that $F_{\pi(n)-1} \equiv 2 - F_{\pi(n)+1} \pmod{p^k}$, or equivalently,

$$2(F_{\pi(n)+1} - 1) = F_{\pi(n)+1} + F_{\pi(n)} + F_{\pi(n)-1} - 2 \equiv F_{\pi(n)} \equiv 2ap^{k-1} \pmod{p^k},$$

which implies that $F_{\pi(n)+1} \equiv 1 + ap^{k-1} \pmod{p^k}$. This establishes the base case of the induction. The inductive step follows from

$$\begin{aligned} F_{(r+1)\pi(n)+1} &\equiv F_{\pi(n)+1}F_{r\pi(n)+1} + F_{\pi(n)}F_{r\pi(n)} \\ &\equiv (1 + ap^{k-1})(1 + arp^{k-1}) \\ &\equiv 1 + a(r+1)p^{k-1} \pmod{p^k}. \end{aligned}$$

Now that we have established (2.4) for $j = 0, 1$, we prove the general case by induction on $j \geq 0$. Suppose the claim is true for all integers less than j . Then

$$\begin{aligned} F_{r\pi(n)+j} &\equiv F_{r\pi(n)+j-2} + F_{r\pi(n)+j-1} \\ &\equiv (F_{j-2} + arp^{k-1}L_{j-2}) + (F_{j-1} + arp^{k-1}L_{j-1}) \\ &\equiv F_j + arp^{k-1}L_j \pmod{p^k}. \end{aligned}$$

The induction step is completed and the lemma follows. $\square$

By Lemma 2.2, $\mu_0(p^k) \geq 1$ for all primes $p \equiv \pm 1 \pmod{5}$ and $k \geq 1$. The following theorem gives a better lower bound.

Theorem 2.6. *Let p be a prime number and let $k \geq 1$. Then every maximal coprime CF-sequence modulo p is a maximal coprime CF-sequence modulo p^k . In particular,*

$$\mu_0(p^k) \geq p^{k-1}\mu_0(p).$$

Proof. Since for $p = 2, 5$, there are no maximal coprime CF-sequences, we can assume that $p \neq 2, 5$. Let t be the largest integer such that $\pi(p^t) = \pi(p)$. Then $\pi(p^k) = p^{k-t}\pi(p)$ for all $k \geq t$ by Lemma 2.4.

Let $f_0 = 1, f_1, \dots$ be a sequence of integers that forms a coprime CF-sequence modulo p . We prove by induction on $k \geq 1$ that it also forms a coprime CF-sequence modulo p^k of length $\pi(p^k)$. The base case is clear. To prove the induction step, suppose $f_i, i \geq 0$, is a coprime CF-sequence modulo p^{k-1} of length $\pi(p^{k-1})$.

Suppose that $f_i \equiv f_j \pmod{p^k}$ for some $i > j \geq 0$, hence $F_i f_1 + F_{i-1} f_0 \equiv F_j f_1 + F_{j-1} f_0 \pmod{p^k}$. We need to show that $\pi(p^k) \mid (i - j)$. From $f_i \equiv f_j \pmod{p^{k-1}}$ and the induction hypothesis, we have $\pi(n) \mid (i - j)$, where $n = p^{k-1}$. If $k \leq t$, then $\pi(p^k) = \pi(p^{k-1}) = \pi(p)$, which implies that $\pi(p^k) \mid (i - j)$ and the claim follows. Thus suppose that $k > t$.

Since $\pi(n) \mid (i - j)$ and $i > j$, there exists an integer $r \geq 1$ such that $i = r\pi(n) + j$. Since $\pi(p^k) = p\pi(p^{k-1})$ for $k > t$, we need to show that $r \equiv 0 \pmod{p}$. For the sake of contradiction, suppose $p \nmid r$. It follows from Lemma 2.5 that

$$F_i \equiv F_j + arp^{k-1}L_j \pmod{p^k}, \tag{2.7}$$

$$F_{i-1} \equiv F_{j-1} + arp^{k-1}L_{j-1} \pmod{p^k}, \tag{2.8}$$

where a is a constant depending only on p and $p \nmid a$. These together with $F_i f_1 + F_{i-1} \equiv F_j f_1 + F_{j-1} \pmod{p^k}$ imply that

$$arp^{k-1}(f_1 L_j + L_{j-1}) \equiv 0 \pmod{p^k},$$

or, equivalently, $f_1 L_j + L_{j-1} \equiv 0 \pmod{p}$. Since $L_k = F_k + 2F_{k-1}$ for all k , we conclude that

$$f_j + 2f_{j-1} \equiv f_1(F_j + 2F_{j-1}) + F_{j-1} + 2F_{j-2} \equiv 0 \pmod{p}.$$

Let $y = f_{j-1}$, then $f_j \equiv -2y \pmod{p}$ and $f_{j+1} \equiv -y \pmod{p}$. By induction on $s \geq 0$:

$$f_{j+s} \equiv -yL_s \pmod{p}.$$

It follows from the identity $L_{\pi(p)-k} \equiv (-1)^k L_k \pmod{p}$ for all k , that

$$f_{j+\pi(p)-2} \equiv -yL_{\pi(p)-2} \equiv -yL_2 \equiv f_{j+2} \pmod{p},$$

contradicting that $f_i, i \geq 0$, is cyclic modulo p of length $\pi(p)$. Thus $r \equiv 0 \pmod{p}$, and the theorem follows. $\square$

It is worth mentioning that the maximality condition in Theorem 2.6 is needed, since for example the Fibonacci sequence $1, 4, \dots$ is a CF-sequence modulo 11 but not modulo 121.

The converse of Theorem 2.6 is true if p is not a Wall-Sun-Sun prime; i.e., if $t(p) = 1$. In this case, Theorem 2.6 can be strengthened as follows.

Theorem 2.7. *Let p be a prime that is not a Wall-Sun-Sun prime, and let $k \geq 1$. Then a Fibonacci sequence is a maximal coprime CF-sequence modulo p if and only if it is a maximal coprime CF-sequence modulo p^k . In particular,*

$$\mu_0(p^k) = p^{k-1} \mu_0(p). \quad (2.9)$$

Proof. *The primes 2 and 5.* Every Fibonacci sequence modulo 2 contains an even term, so there are no coprime sequences modulo 2^k . For $p = 5$, a coprime CF-sequence modulo 5^k has $\pi(5^k) = 4 \times 5^k$ pairwise distinct terms if it is maximal, all relatively prime with 5; but there are only $\varphi(5^k) = 4 \times 5^{k-1}$ congruence classes modulo 5^k that are relatively prime with 5. Henceforth $p \neq 2, 5$.

The case $k = 1$ is trivial, and the implication “mod $p \Rightarrow$ mod p^k ” is Theorem 2.6. It remains to prove the converse for $k \geq 2$.

Let $f_0 = 1, f_1 = y, \dots$ be a maximal coprime CF-sequence modulo p^k , and set

$$L = \pi(p^k) = p^{k-1} \pi(p).$$

The terms $f_0, \dots, f_{L-1}$ are pairwise incongruent modulo p^k .

Reduce the sequence modulo p . Since $p \nmid f_i$ for all i , it is a coprime Fibonacci sequence modulo p with $f_0 = 1$. Let d be its least period modulo p . Then $d \mid \pi(p)$, and $d \mid L$ because L is a period modulo p^k , hence modulo p .

Fix a residue r modulo p that occurs in the sequence, and let $m_r \geq 1$ be the number of indices $0 \leq i < d$ with $f_i \equiv r \pmod{p}$. Among $f_0, \dots, f_{L-1}$, which consist of L/d

full periods modulo p , the class r occurs exactly $(L/d)m_r$ times. These terms are pairwise incongruent modulo p^k , and exactly p^{k-1} residues modulo p^k reduce to r . Therefore

$$\frac{L}{d}m_r \leq p^{k-1} \iff \frac{\pi(p)}{d}m_r \leq 1.$$

As $d \mid \pi(p)$ and $m_r \geq 1$, this forces $d = \pi(p)$ and $m_r = 1$. In particular $f_0, \dots, f_{\pi(p)-1}$ are pairwise incongruent modulo p , so the sequence is a CF-sequence modulo p of length $\pi(p)$. It is coprime, hence a maximal coprime CF-sequence modulo p .

To prove (2.9), let $\Lambda_0(n)$ denote the set of integers y modulo n such that the Fibonacci sequence $1, y, \dots$ is a maximal coprime CF-sequence modulo n . One has $\mu_0(n) = |\Lambda_0(n)|$. Let $\rho: \mathbb{Z}_{p^k} \rightarrow \mathbb{Z}_p$ be the reduction modulo p . We have shown so far that $\Lambda_0(p^k) = \rho^{-1}(\Lambda_0(p))$. Every set $\rho^{-1}(i)$ for $1 \leq i \leq p-1$ has p^{k-1} elements, so $|\Lambda_0(p^k)| = p^{k-1}|\Lambda_0(p)|$, and the claim follows. $\square$

The hypothesis $t(p) = 1$ cannot be dropped from Theorem 2.7. Suppose $p \equiv \pm 1 \pmod{5}$ has $t(p) \geq 2$, and let α, β be the roots of $x^2 - x - 1$ modulo p with $\text{ord}_p(\alpha)$ even and $h = \text{ord}_p(\beta)$ odd. Let α_2, β_2 be their lifts to $\mathbb{Z}_{p^2}$, so $\alpha_2\beta_2 = -1$, $\pi(p^2) = \pi(p) = 2h$, and $\text{ord}_{p^2}(\beta_2) = h$. Choose $y = \beta_2 + c$ with $p \mid c$ and $p^2 \nmid c$. Then consider the Fibonacci sequence

$$f_i \equiv \beta_2^i + \gamma(\alpha_2^i - \beta_2^i) \pmod{p^2},$$

where $\gamma = \frac{c}{\alpha_2 - \beta_2}$. If $f_i \equiv f_j \pmod{p^2}$, reduction modulo p gives $j \equiv i \pmod{h}$, so $j \equiv i$ or $i \pm h \pmod{2h}$. But $\alpha_2^h \equiv -1$ and $\beta_2^h \equiv 1$ give $f_i - f_{i+h} \equiv 2\gamma\alpha_2^i \not\equiv 0 \pmod{p^2}$. Hence $f_0, \dots, f_{2h-1}$ are distinct, and the sequence is a maximal coprime CF-sequence modulo p^2 , while its reduction modulo p is the geometric sequence β^i of length $h < \pi(p)$. No Wall–Sun–Sun prime is known, so this scenario is hypothetical.

The next lemma relates CF-sequences modulo m, n to those modulo mn .

Lemma 2.8. *Let m, n be relatively prime positive integers. Then*

$$\mu(mn) \geq \mu(m)\mu(n); \tag{2.10}$$

$$\mu_0(mn) \geq \mu_0(m)\mu_0(n). \tag{2.11}$$

Proof. For an integer x , let $\Lambda(x)$ denote the set of integers $0 \leq y < x$ such that the Fibonacci sequence $1, y, \dots$ is a coprime CF-sequence modulo x . One has $\mu(x) = |\Lambda(x)|$. We define a map

$$\theta: \Lambda(m) \times \Lambda(n) \rightarrow \Lambda(mn),$$

as follows. Given $a \in \Lambda(m)$ and $b \in \Lambda(n)$, by the Chinese Remainder Theorem, there exists a unique integer $c \in \{0, \dots, mn-1\}$ such that $c \equiv a \pmod{m}$ and $c \equiv b \pmod{n}$. We claim that $c \in \Lambda(mn)$. Let $f_i, i \geq 0$ be the Fibonacci sequence with initial terms $f_0 = 1$ and $f_1 = c$. Since the sequence $f_i, i \geq 0$ reduces to $1, a, \dots$ modulo m , it is a CF-sequence modulo m , and so $\gcd(f_i, m) = 1$; similarly $\gcd(f_i, n) = 1$ for all $i \geq 0$. Since $\gcd(m, n) = 1$, it follows that $\gcd(f_i, mn) = 1$.

To see that $f_i, i \geq 0$, is a CF-sequence modulo mn , suppose that $f_i \equiv f_j \pmod{mn}$, and we prove that $f_{i+1} \equiv f_{j+1} \pmod{mn}$. It follows from $f_i \equiv f_j \pmod{mn}$ that

$f_i \equiv f_j \pmod{m}$, and since $f_i, i \geq 0$ is a CF-sequence modulo m , we have $f_{i+1} \equiv f_{j+1} \pmod{m}$. Similarly, $f_{i+1} \equiv f_{j+1} \pmod{n}$. Again, since $\gcd(m, n) = 1$, we conclude that $f_{i+1} \equiv f_{j+1} \pmod{mn}$.

The map θ is injective, since c is uniquely determined by a, b . Therefore, $|\Lambda(mn)| \geq |\Lambda(m)| \cdot |\Lambda(n)|$, and (2.10) follows.

To prove (2.11), it is sufficient to show that if $1, a, \dots$ is maximal modulo m and $1, b, \dots$ is maximal modulo n , then $1, c, \dots$ is maximal modulo mn . This follows from the fact that the period of a Fibonacci sequence modulo mn is the least common multiple of the periods modulo m and n . In this case $\text{lcm}(\pi(m), \pi(n)) = \pi(mn)$, and so (2.11) follows from the injectivity of θ . $\square$

We note that the map θ is not generally surjective. For example, with $m = 5$ and $n = 29$, one has $8 \in \Lambda(5 \times 29)$, since $1, 8, 9, \dots, 139, 7$ is a coprime CF-sequence modulo $145 = 5 \times 29$. However, there are no values $(a, b) \in \Lambda(5) \times \Lambda(29)$ with $\theta(a, b) = 8$, since the sequence reduced modulo 29 is not cyclic.

As another example, with $m = 17$ and $n = 19$, we have the maximal coprime CF-sequence $1, 15, \dots$ modulo 17×19 while its reduction modulo 17 is not a coprime CF-sequence.

The following theorem follows from Theorems 2.6 and 2.8.

Theorem 2.9. *Let $n = p_1^{k_1} \cdots p_m^{k_m}$, where $p_1, \dots, p_m$ are distinct primes with $p_i \equiv \pm 1 \pmod{5}$ and $k_i \geq 1$ for all $1 \leq i \leq m$. Then*

$$\mu(n) \geq \frac{2^s n}{p_1 \cdots p_m} \geq 2^m, \quad (2.12)$$

$$\mu_0(n) \geq \frac{n}{p_1 \cdots p_m}, \quad (2.13)$$

where $s = \#\{1 \leq i \leq m : k_i = 1\}$.

Proof. By Lemma 2.8, we have

$$\mu(n) \geq \mu(p_1^{k_1}) \cdots \mu(p_m^{k_m}).$$

By Theorem 2.6, we have $\mu(p_i^{k_i}) \geq \mu_0(p_i^{k_i}) \geq p_i^{k_i-1} \mu_0(p_i) \geq p_i^{k_i-1}$. The inequality in (2.12) then follows from this and $\mu(p_i) \geq 2$. The proof of (2.13) is similar. $\square$

3. Integers with no Coprime CF-sequences

The next lemma gives examples of integers that do not admit coprime CF-sequences.

Lemma 3.1. *Let n be an integer with a prime divisor p such that $p = 2$ or $p \equiv 3, 7 \pmod{20}$ and $\pi(p) = 2(p+1)$. Then $\mu(n) = 0$.*

Proof. If a prime p satisfies the hypothesis of the theorem, then every Fibonacci sequence contains a multiple of p [1]. Therefore, if an integer n has such a prime divisor p , every Fibonacci sequence contains a term that is not relatively prime with n , hence $\mu(n) = 0$. $\square$

The primes p satisfying the hypothesis of Lemma 3.1 are primes that divide at least

one term in every Fibonacci sequence. They are listed as [14]

$$A000057 : 2, 3, 7, 23, 43, 67, 83, 103, 127, 163, 167, 223, 227, 283, \dots$$

It is not known if this sequence is infinite, but assuming the generalized Riemann hypothesis, the asymptotic density of these primes is positive. To be precise, let $\pi(x)$ denote the number of primes less than or equal to x . A set of primes P is said to have positive asymptotic density if

$$\lim_{x \rightarrow \infty} \frac{\pi_P(x)}{\pi(x)} > 0,$$

where $\pi_P(x) = \#\{p \in P : p \leq x\}$.

The natural density of a set A of positive integers is defined as

$$d(A) = \lim_{x \rightarrow \infty} \frac{\#\{n \in A : n \leq x\}}{x},$$

where $\#X$ denotes the number of elements of X .

Lemma 3.2. *Let P be a set of primes with positive asymptotic density. Then the natural density of integers n with the property that $p \nmid n$ for all $p \in P$ is zero.*

Proof. Let $P = \{p_1, p_2, \dots\}$, where $p_1 < p_2 < \dots$, and let

$$S(x) = \{n \in \mathbb{N} : n \leq x \text{ and } p \nmid n \text{ for all } p \in P\}.$$

We need to show that $\lim_{x \rightarrow \infty} \#S(x)/x = 0$.

For each $k \geq 1$, the proportion of integers modulo $p_1 \cdots p_k$ that are relatively prime with $p_1, \dots, p_k$ is given by

$$\frac{\phi(p_1 \cdots p_k)}{p_1 \cdots p_k} = \prod_{i=1}^k \left(1 - \frac{1}{p_i}\right).$$

Let $x = p_1 \cdots p_k M + y$, where $M \geq 0$ and $0 \leq y < p_1 \cdots p_k$. Then

$$S(x) \subseteq S(p_1 \cdots p_k M) \cup (\mathbb{N} \cap (p_1 \cdots p_k M, x]).$$

It follows that

$$\#S(x) \leq \#S(p_1 \cdots p_k M) + x - p_1 \cdots p_k M \leq p_1 \cdots p_k M \prod_{i=1}^k \left(1 - \frac{1}{p_i}\right) + p_1 \cdots p_k,$$

and so

$$\frac{\#S(x)}{x} \leq \prod_{i=1}^k \left(1 - \frac{1}{p_i}\right) + \frac{p_1 \cdots p_k}{x}.$$

It follows from this and the inequality $1 - t \leq e^{-t}$ for all $t \geq 0$ that

$$\limsup_{x \rightarrow \infty} \frac{\#S(x)}{x} \leq \prod_{i=1}^k \left(1 - \frac{1}{p_i}\right) \leq \prod_{i=1}^k \exp\left(-\frac{1}{p_i}\right) \leq \exp\left(-\sum_{i=1}^k \frac{1}{p_i}\right),$$

for all $k \geq 1$, which implies that

$$\limsup_{x \rightarrow \infty} \frac{\#S(x)}{x} \leq \exp\left(-\sum_{i=1}^{\infty} \frac{1}{p_i}\right).$$

Therefore, to prove the lemma, it is sufficient to show that $\sum_{i=1}^{\infty} 1/p_i = \infty$.

By Abel's summation formula, we have

$$\sum_{\substack{p \in P \\ p \leq x}} \frac{1}{p} = \frac{\pi_P(x)}{x} + \int_2^x \frac{\pi_P(t)}{t^2} dt. \quad (3.1)$$

By the assumption in the lemma, $\pi_P(t)/\pi(t) > \delta > 0$ for t large enough. On the other hand, by the prime number theorem $\pi(t) > t/(2 \ln t)$ as $t \rightarrow \infty$. It follows that $\pi_P(t)/t^2 > \delta/(2t \ln t)$ for t large enough. Since $\int_2^{\infty} dt/(t \ln t)$ is divergent, by comparison, the integral in (3.1) is divergent as well, and the lemma follows. $\square$

Theorem 3.3. *(Conditional on GRH) The natural density of positive integers that admit a coprime CF-sequence is zero.*

Proof. Since every integer n that satisfies the conditions of Lemma 3.1 has $\mu(n) = 0$ and the natural density of these integers is zero by Lemma 3.2, the density of the set of integers with $\mu(n) > 0$ is zero. $\square$

4. CF-lengths Modulo Integers

A coprime CF-length modulo n is a number k such that there exists a coprime CF-sequence of length k modulo n . In this section, we investigate the possible lengths of CF-sequences.

The following lemma gives a complete description of odd-length coprime CF-sequences modulo integers.

Lemma 4.1. *Suppose that $f_0, f_1, \dots$ is a coprime Fibonacci sequence of odd period modulo an integer n . Then every prime factor of n satisfies $p \equiv \pm 1 \pmod{5}$ and f_1/f_0 is the unique root of $x^2 - x - 1 = 0$ modulo n with $\text{ord}_n(f_1/f_0) = \pi(n)/2$. In particular, every coprime Fibonacci sequence with odd period modulo an integer n is a coprime CF-sequence modulo n induced by a Fibonacci root modulo n and is unique up to multiplication.*

Proof. We first note that $5 \nmid n$, since every Fibonacci sequence modulo 5 has even period. Similarly $2 \nmid n$, since every Fibonacci sequence contains an even number, hence it would not be a coprime sequence modulo n if n is even. Let d be the least period

Table 4.1. Lengths of a coprime CF-sequence modulo $n < 100$

Integer n	coprime CF-length(s)
5	4
11	5, 10
19	9, 18
29	7, 14
31	15, 30
41	40
55	20
59	29, 58
61	60
71	35, 70
79	39, 78
89	44
95	36

length of a coprime Fibonacci sequence $f_i, i \geq 0$, modulo n . Then by (1.1), we have

$$\begin{cases} F_d f_1 + (F_{d-1} - 1)f_0 \equiv 0 \pmod{n}; \\ (F_{d+1} - 1)f_1 + F_d f_0 \equiv 0 \pmod{n}. \end{cases} \quad (4.1)$$

From this system of equations, we have

$$F_d^2 \equiv (F_{d-1} - 1)(F_{d+1} - 1) \equiv F_{d-1}F_{d+1} - F_{d-1} - F_{d+1} + 1 \pmod{n},$$

and so

$$F_{d-1} + F_{d+1} \equiv F_{d-1}F_{d+1} - F_d^2 + 1 \equiv (-1)^d + 1 \equiv 0 \pmod{n}. \quad (4.2)$$

Let $y = F_{d-1}$, and so $F_{d+1} \equiv -y \pmod{n}$ and $F_d = F_{d+1} - F_{d-1} \equiv -2y \pmod{n}$. From the identity $F_{d-1}F_{d+1} = F_d^2 + (-1)^d$, we conclude that $-y^2 \equiv (-2y)^2 - 1 \pmod{n}$, or equivalently, $5y^2 \equiv 1 \pmod{n}$, hence 5 is a quadratic residue modulo n . It follows that 5 is a quadratic residue modulo every prime factor p of n , or equivalently, $p \equiv \pm 1 \pmod{5}$.

It follows from (4.1) that $-2yf_1 + (y - 1)f_0 \equiv 0 \pmod{p}$, and so $f_1/f_0 \equiv (y - 1)/(2y) \pmod{p}$. A simple algebraic calculation shows that $5y^2 \equiv 1 \pmod{p}$ implies $(y - 1)/(2y)$ is a root of $x^2 - x - 1$ modulo n , hence f_1/f_0 is a root of $x^2 - x - 1$ modulo every prime power factor p^m of n . Since the least period of $f_i, i \geq 0$, modulo n is odd, $\text{ord}_{p^m}(f_1/f_0)$ is odd, and by Lemma 2.2, it equals $\pi(p^m)/2$. Therefore, if $n = p_1^{m_1} \cdots p_k^{m_k}$ is the prime factorization of n , we have

$$\text{ord}_n(f_1/f_0) = \text{lcm}(\text{ord}_{p_1^{m_1}}(f_1/f_0), \dots, \text{ord}_{p_k^{m_k}}(f_1/f_0)) \quad (4.3)$$

$$= \text{lcm}(\pi(p_1^{m_1})/2, \dots, \pi(p_k^{m_k})/2) \quad (4.4)$$

$$= \frac{1}{2} \text{lcm}(\pi(p_1^{m_1}), \dots, \pi(p_k^{m_k})) \quad (4.5)$$

$$= \frac{1}{2} \pi(n). \quad (4.6)$$

It follows that the only coprime Fibonacci sequence modulo n with odd period is equivalent to the geometric sequence $1, x, x^2, \dots$ where $x^2 - x - 1 \equiv 0 \pmod{n}$ with length $\text{ord}_n(x) = \pi(n)/2$. Note that such x modulo n is unique if it exists, since $x \pmod{p_i^{m_i}}$ is unique for each $1 \leq i \leq k$. $\square$

The uniqueness property in Lemma 4.1 does not hold for coprime Fibonacci sequences with even period. For example, there are 112 non-equivalent coprime CF-sequences modulo 199, all of length 22. Lemma 4.1 fails for even-length Fibonacci sequences in one additional way: not every coprime Fibonacci sequence with even least period is necessarily a CF-sequence. For example, modulo 13 the Fibonacci sequence $1, 7, \dots$ is a coprime Fibonacci sequence of least period 28, hence it is not a CF-sequence.

Lemma 4.2. *Let $n = p_1^{m_1} \cdots p_k^{m_k}$, where $p_i \neq 5$ are distinct primes and $m_i \geq 1$ for all $1 \leq i \leq k$. If $f_0, f_1, \dots$ is a coprime Fibonacci sequence modulo n of even length d , then $d = \pi(n)$.*

Proof. Equation (4.2) turns into $F_{d-1} + F_{d+1} \equiv 2 \pmod{n}$. Letting $y = F_{d-1}$, we have $F_{d+1} \equiv 2 - y \pmod{n}$ and $F_d \equiv 2 - 2y \pmod{n}$.

By reduction modulo $p_i^{m_i}$, we conclude from (4.1) that $(2 - 2y)f_1 + (y - 1)f_0 \equiv 0 \pmod{n}$ and $(1 - y)f_1 + (2 - 2y)f_0 \equiv 0 \pmod{n}$, and so

$$(f_0 - 2f_1)(y - 1) \equiv 0 \pmod{p_i^{m_i}} \quad (4.7)$$

$$(2f_0 + f_1)(y - 1) \equiv 0 \pmod{p_i^{m_i}} \quad (4.8)$$

Since $p_i \neq 5$, either $p_i \nmid (f_0 - 2f_1)$ or $p_i \nmid (2f_0 + f_1)$, and so $y \equiv 1 \pmod{p_i^{m_i}}$, which in turn implies that $F_d \equiv 0 \pmod{p_i^{m_i}}$ and $F_{d+1} \equiv 1 \pmod{p_i^{m_i}}$ for all $1 \leq i \leq k$. Therefore, $F_d \equiv 0 \pmod{n}$ and $F_{d+1} \equiv 1 \pmod{n}$, and so $\pi(n) \mid d$. Since $d \mid \pi(n)$ also, we conclude that $d = \pi(n)$. $\square$

Lemma 4.2 fails if $5 \mid n$. For example, for $n = 145$, the coprime Fibonacci sequence $1, 8, \dots$ has length $28 \neq \pi(145)$.

Lemma 4.3. *Let $s \geq 1$. If $i \equiv 0 \pmod{4 \times 5^{s-1}}$, then $F_i + 2F_{i-1} \equiv 2 \pmod{5^s}$.*

Proof. Let $i = 4r \times 5^{s-1} = r\pi(n)$, where $n = 5^{s-1}$. By Lemma 2.5, we have

$$F_i \equiv 3r \times 5^{s-1} \pmod{5^s}, \quad F_{i-1} \equiv 1 + r \times 5^{s-1} \pmod{5^s},$$

which imply that $F_i + 2F_{i-1} \equiv 2 \pmod{5^s}$. $\square$

Lemma 4.4. *Let $n = 5^s m$ be a positive integer such that $s \geq 1$ and $5 \nmid m$. Let d be the least period of a coprime Fibonacci sequence modulo n . Then*

$$d = \begin{cases} \pi(n) & \text{if } 5^s \mid \pi(m); \\ \pi(n)/5 & \text{if } 5^s \nmid \pi(m). \end{cases}$$

Proof. Let $f_0, f_1, \dots$ be a coprime Fibonacci sequence modulo n with length d . By Lemma 4.1, d is even.

Letting $y = F_{d-1}$, from (4.7) and (4.8), we have $y \equiv 1 \pmod{m}$. Additionally, since either $25 \nmid (f_0 - 2f_1)$ or $25 \nmid (2f_0 + f_1)$, we have $y \equiv 1 \pmod{5^{s-1}}$. It follows that $F_d \equiv 0 \pmod{n/5}$ and $F_{d+1} \equiv 1 \pmod{n/5}$, hence $\pi(n/5) \mid d$.

Let $e = \pi(n/5)$. For $s \geq 2$, we have

$$\pi(n) = \text{lcm}(4 \times 5^s, \pi(m)) = \begin{cases} \text{lcm}(4 \times 5^{s-1}, \pi(m)) & \text{if } 5^s \mid \pi(m); \\ 5 \text{lcm}(4 \times 5^{s-1}, \pi(m)) & \text{if } 5^s \nmid \pi(m). \end{cases}$$

Case 1. Suppose that $s > 1$ and $5^s \mid \pi(m)$. In this case $\pi(n) = \pi(m) = e$, hence $d \mid e$. This together with $e \mid d$ implies that $d = e = \pi(n)$.

Case 2. Suppose that $s > 1$ and $5^s \nmid \pi(m)$. In this case $\pi(n) = 5\pi(m) = 5e$, hence $d \mid 5e$, and so $d = e$ or $d = 5e$. To prove $d = e$, we need to show that $f_e \equiv f_0 \pmod{5^s}$ and $f_{e+1} \equiv f_1 \pmod{5^s}$. Write $f_0 = 2f_1 + 5\ell$. Since $F_{e-1} \equiv 1 \pmod{5^{s-1}}$ and $f_0 \equiv 2f_1 \pmod{5}$, we have

$$f_e = F_e f_1 + (F_{e-1} - 1)f_0 \equiv (F_e + 2(F_{e-1} - 1))f_1 \equiv 0 \pmod{5^s},$$

by Lemma 4.3.

Case 3. Suppose that $s = 1$, then

$$\pi(n) = \text{lcm}(20, \pi(m)) = \begin{cases} \pi(m) & \text{if } 20 \mid \pi(m); \\ 2\pi(m) & \text{if } 4 \nmid \pi(m) \text{ and } 5 \mid \pi(m); \\ 5\pi(m) & \text{if } 4 \mid \pi(m) \text{ and } 5 \nmid \pi(m); \\ 10\pi(m) & \text{if } 4 \nmid \pi(m) \text{ and } 5 \nmid \pi(m). \end{cases}$$

We claim that $g = \text{lcm}(4, e) = e$ or $2e$ is a period of $f_i, i \geq 0$. This follows from the fact that $F_i + 2(F_{i-1} - 1) \equiv 0 \pmod{5}$ for all $i \equiv 0 \pmod{4}$. It follows that $d \mid g$. This together with $e \mid d$ implies that $d = e = g$ (if $e \equiv 0 \pmod{4}$) or $d = 2e = g$ (if $e \equiv 2 \pmod{4}$). Now $\pi(n) = \text{lcm}(5, g)$. If $5 \mid e$, then $\pi(n) = g = d$ and if $5 \nmid e$, then $\pi(n) = 5g = 5d$. $\square$

The following theorem follows from Lemmas 4.1, 4.2, and 4.4.

Theorem 4.5. *Let $n = 5^s m$ be a positive integer, where $s \geq 0$ and $5 \nmid m$. Let d be the length of a coprime Fibonacci sequence $f_0, f_1, \dots$ modulo n .*

- i) *If d is odd, then $d = \pi(n)/2$, every prime factor of n equals $\pm 1 \pmod{5}$, and the sequence is equivalent to the geometric sequence $1, x, x^2, \dots$ where x is the only root of $x^2 - x - 1$ modulo n with odd multiplicative order modulo n .*
- ii) *If d is even and $5^s \mid \pi(m)$, then $d = \pi(n)$.*
- iii) *If d is even and $5^s \nmid \pi(m)$, then $d = \pi(n)/5$.*

In particular, there exist at most two possible lengths for coprime CF-sequences modulo n , and if there are two lengths, one is twice the other.

Corollary 4.6. *Let $p \neq 2, 5$ be a prime number.*

- i) *If $p \equiv \pm 1 \pmod{5}$ and $\pi(p) \equiv 0 \pmod{4}$, then every CF-sequence modulo p is maximal; i.e., $\mu_0(p) = \mu(p)$.*
- ii) *If $p \equiv \pm 1 \pmod{5}$ and $\pi(p) \equiv 2 \pmod{4}$, then every CF-sequence modulo p is maximal except one; i.e., $\mu_0(p) = \mu(p) - 1$.*
- iii) *If $p \equiv \pm 2 \pmod{5}$, then every CF-sequence modulo p is maximal; i.e., $\mu_0(p) = \mu(p)$.*

5. CF-sequences in Finite Groups

Example 5.1. Let $G = \mathbb{Z}_7 \times \mathbb{Z}_7$. Let $a_0 = (1, 1)$ and $a_1 = (2, 3)$, then we have

$$(1, 1), (2, 3), (3, 4), (5, 0), \dots, (0, 6), (1, 2), (1, 1), (2, 3) \dots$$

the cyclic fibonacci sequence in G . More generally, every group of the form $G = \mathbb{Z}_n \times \mathbb{Z}_n$ admits a CF-sequence, since the Fibonacci sequence in G starting with $g_0 = (0, 1)$ and $g_1 = (1, 0)$ is always cyclic.

Theorem 5.2. Every non-cyclic abelian group admits a CF-sequence.

Proof. Let $G \cong \mathbb{Z}_{n_1} \times \dots \times \mathbb{Z}_{n_k}$ be the invariant factor decomposition of group G , where $n_i \mid n_{i+1}$ for $1 \leq i \leq k-1$. In particular, $\mathbb{Z}_{n_1} \times \mathbb{Z}_{n_1}$ is a subgroup of G , hence it is sufficient to prove the claim in the case $G = \mathbb{Z}_n \times \mathbb{Z}_n$.

Let $g_0, g_1, \dots$ be the Fibonacci sequence in G with initial terms $g_0 = (0, 1)$ and $g_1 = (1, 0)$. We need to show that if $g_i \equiv g_j$, then $g_{i+1} \equiv g_{j+1}$. By induction on $k \geq 0$, one has

$$g_k = (F_k, F_{k-1}), \quad k \geq 0.$$

Now, if $g_i \equiv g_j$ for some $i, j \geq 0$, then

$$(F_i, F_{i-1}) = (F_j, F_{j-1}),$$

and so $F_i \equiv F_j \pmod{n}$ and $F_{i-1} \equiv F_{j-1} \pmod{n}$. It follows that $F_{i+1} \equiv F_{j+1} \pmod{n}$ (and also $\pi(n) \mid (i - j)$), hence $g_{i+1} = g_{j+1}$. We conclude that $g_0, g_1, \dots$ is a cyclic Fibonacci sequence with period $\pi(n)$. $\square$

Example 5.3. Both $\mathbb{Z}_2 \times \mathbb{Z}_2$ and $\mathbb{Z}_3 \times \mathbb{Z}_3$ admit reduced CF-sequences:

$$\begin{aligned} \mathbb{Z}_2 \times \mathbb{Z}_2 : & (0, 1), (1, 0), (1, 1), \dots; \\ \mathbb{Z}_3 \times \mathbb{Z}_3 : & (0, 1), (1, 0), (1, 1), (2, 1), (0, 2), (2, 0), (2, 2), (1, 2), \dots \end{aligned}$$

Let G be a group that admits a reduced Fibonacci sequence and let N be a proper normal subgroup of G . The projection $G \mapsto G/N$ maps a reduced Fibonacci sequence to a complete Fibonacci sequence (a sequence that includes every element of G/N). It follows from Theorem 1.5 that G/N is a cyclic group of order $n = 5^k m$, where $k \geq 0$ and $m \in \{1, 2, 4, 6, 7, 14\} \cup \{3^j : j \geq 0\}$.

Theorem 5.4. The only non-cyclic abelian groups that admit reduced Fibonacci sequences are $\mathbb{Z}_2 \times \mathbb{Z}_2$ and $\mathbb{Z}_3 \times \mathbb{Z}_3$.

Proof. Let $G \cong \mathbb{Z}_{n_1} \times \mathbb{Z}_{n_2} \times \dots \times \mathbb{Z}_{n_k}$ be the invariant factor representation of a non-cyclic abelian group that admits a reduced CF-sequence. The quotient groups G/N must be cyclic for every proper normal subgroup N of G . It follows immediately that $k = 2$ and $n_1 = n_2 = p$, where p is a prime number, hence $G \cong \mathbb{Z}_p \times \mathbb{Z}_p$ and the quotient group $\mathbb{Z}_p \cong G/\mathbb{Z}_p$ must also admit a complete Fibonacci sequence. It again follows from Theorem 1.5 that $p \in \{2, 3, 5, 7\}$. A simple inspection shows that only in the cases $p = 2, 3$ we obtain a reduced Fibonacci sequence. $\square$

Next, we discuss CF-sequences in semidirect products of cyclic groups.

Definition 5.5. Let m, n, a be positive integers such that $a^n \equiv 1 \pmod{m}$. The semidirect product $\mathbb{Z}_m \rtimes \mathbb{Z}_n$ is defined by

$$(x_1, y_1) * (x_2, y_2) = (x_1 + a^{y_1} x_2, y_1 + y_2).$$

Example 5.6. The following is a reduced (but not cyclic) Fibonacci sequence in $\mathbb{Z}_3 \rtimes \mathbb{Z}_2$ (with $a = 2$):

$$(0, 1), (1, 0), (2, 1), (0, 1), (2, 0), (1, 1), \dots$$

It is conjectured that there are no reduced Fibonacci sequences in a non-abelian group of order > 6 . This conjecture is supported by data from Table 5.1.

Table 5.1. Coverage and cyclic Fibonacci sequences in small non-abelian groups. Coverage is the number of distinct elements visited over one period of a Fibonacci sequence, maximized over all starting pairs $(g_0, g_1) \in G \times G$ (exhaustive search); it need not come from a cyclic sequence. “Reduced?” asks whether some sequence (cyclic or not) attains coverage $|G| - 1$ while omitting the identity. “CF lengths” lists the periods of the *cyclic* Fibonacci sequences in G , excluding the trivial constant sequence at the identity (length 1).

Group	$ G $	Max coverage	Reduced?	CF-sequence?	CF lengths
$S_3 \cong D_3$	6	5/6	Yes	No	—
D_4	8	5/8	No	Yes	3
Q_8 (Dic ₂)	8	4/8	No	Yes	3
D_5	10	5/10	No	Yes	4
D_6	12	6/12	No	Yes	3
Dic ₃ (Q_{12})	12	6/12	No	No	—
A_4	12	10/12	No	Yes	3
D_7	14	7/14	No	No	—
D_8	16	6/16	No	Yes	3
Q_{16} (Dic ₄)	16	6/16	No	Yes	3
D_9	18	9/18	No	No	—
D_{10}	20	10/20	No	Yes	3, 4
Q_{20} (Dic ₅)	20	10/20	No	Yes	4
$\mathbb{Z}_7 \rtimes \mathbb{Z}_3$	21	16/21	No	No	—
S_4	24	13/24	No	Yes	3

Example 5.7. The Fibonacci sequence in $G = \mathbb{Z}_p \rtimes \mathbb{Z}_5$ with $f_0 = (0, 1)$ and $f_1 = (1, 3)$ is a CF-sequence of length $4p$ in G . The semidirect product in G is given by

$$(a, b) * (c, d) = (a + t^b c, b + d).$$

It is straightforward to show by induction on $n \geq 0$ that

$$f_{4n+j} = \begin{cases} (-nt^2 - n, 1) & \text{if } j = 0; \\ (-nt^2 + 1, 3) & \text{if } j = 1; \\ (nt^4 + (n+1)t, 4) & \text{if } j = 2; \\ ((n+1)t^4 + 1, 2) & \text{if } j = 3. \end{cases}$$

To prove cyclicity, suppose $f_i = f_j$. First, suppose that $i = 4n$ for some integer n , hence $j = 4m$ for some integer m . It follows that $f_{4n} = f_{4m}$, we must have $-nt^2 - n = -mt^2 - m \pmod{p}$, which is possible only if $n = m \pmod{p}$. Similarly this applies for other indices. It follows that $f_0, f_1, \dots$ is a coprime CF-sequence of length $4p$ in G . It is coprime, since every proper subgroup of G contains less than $4p$ elements.

Next, we discuss coprime CF-sequences in abelian groups.

Example 5.8. Let $G \cong \mathbb{Z}_{2^k} \times \mathbb{Z}_{7 \times 2^k}$, $k \geq 5$. Then the Fibonacci sequence $g_i, i \geq 0$, with starting terms $g_0 = (0, 1)$ and $g_1 = (1, 1)$ is a coprime CF-sequence in G . More generally, if $m \mid n$ and $\pi(m) = \pi(n)$, then the Fibonacci sequence with starting terms $(0, 1), (1, 1)$ is a coprime CF-sequence in $\mathbb{Z}_m \times \mathbb{Z}_n$. In this case $\pi(7) = 16$, and so for $k \geq 5$:

$$\pi(7 \times 2^k) = \text{lcm}(\pi(7), \pi(2^k)) = \text{lcm}(16, 3 \times 2^{k-1}) = 3 \times 2^{k-1} = \pi(2^k).$$

Example 5.9. Let $G \cong \mathbb{Z}_2 \times \mathbb{Z}_{2^k}$, $k \geq 2$. Then there are no coprime CF-sequences in G .

Proof: On the contrary, suppose $g_0, g_1, \dots$ is a coprime CF-sequence in G . Without loss of generality, we can assume $g_0 = (0, \alpha)$ and $g_1 = (1, \beta)$. Since $g_0, g_1, \dots$ is a coprime CF-sequence, at least one of α or β is odd, otherwise the sequence is contained in the subgroup $\mathbb{Z}_2 \times \mathbb{Z}_{2^{k-1}}$. Then

$$g_i = (F_i, \beta F_i + \alpha F_{i-1}).$$

For $i \equiv 0 \pmod{3}$ and $k \geq 5$, we have [8]:

$$F_{i+3 \times 2^{k-2}} \equiv F_i \pmod{2^k}; \quad (5.1)$$

$$F_{i \pm 1 + 3 \times 2^{k-2}} \equiv F_{i \pm 1} + 2^{k-1} \pmod{2^k}; \quad (5.2)$$

It follows that if α is even and β is odd, then $g_{3 \times 2^{k-2}} = g_0$, while $g_{1+3 \times 2^{k-2}} \neq g_1$. Similarly, if α is odd and β is even, then $g_{3 \times 2^{k-2}} \neq g_0$, while $g_{1+3 \times 2^{k-2}} = g_1$. Finally, if both α and β are odd, then $g_{2+3 \times 2^{k-2}} = g_2$, while $g_{1+3 \times 2^{k-2}} \neq g_1$. In each of these cases, the sequence is not cyclic, and the claim follows.

6. Some Open Problems

Not every coprime CF-sequence is induced by a Fibonacci root. As shown in Table 6.1, there are primes (including primes with $p \equiv \pm 2 \pmod{5}$) that admit noticeably more non-equivalent coprime CF-sequences than the at most two induced by the Fibonacci roots. This raises the following open questions: Are there infinitely many primes p for which $\mu(p) > 2$? More generally, is $\limsup_{p \rightarrow \infty} \mu(p) = \infty$?

Table 6.1. Primes < 1000 with more than two CF-sequences

p	199	211	233	281	421	461	521	557	869	911
$\mu(p)$	112	44	104	86	170	140	392	186	236	352

A prime $p \equiv \pm 2 \pmod{5}$, for which Fibonacci roots modulo p do not exist, can

nevertheless admit a coprime CF-sequence. For example, modulo 233, the sequence

$$1, 7, 8, 15, 23, 38, 61, 99, 160, \dots, 70, 190, 27, 217, 11, 228, 6,$$

is a coprime CF-sequence. Table 6.2 lists the first few primes $p \equiv \pm 2 \pmod{5}$ that admit a coprime CF-sequence $f_0 = 1, f_1, \dots$ together with one corresponding choice of f_1 . This leads to the following open question: Are there infinitely many primes $p \equiv \pm 2 \pmod{5}$ for which $\mu(p) > 0$?

Table 6.2. The first nine primes $\equiv \pm 2 \pmod{5}$ with CF-sequences

p	233	557	1087	1103	1427	1597	2207	2417	3167
f_1	7	9	12	9	11	7	7	7	7

There remain many open questions concerning Fibonacci sequences in finite groups. We conclude by highlighting a few of them:

- Let $\lambda(G)$ denote the maximum possible length of a Fibonacci sequence in a non-cyclic finite group G . What is $\limsup \lambda(G)/G$ as $|G| \rightarrow \infty$?
- Which finite abelian groups admit a coprime CF-sequence?
- Which finite groups admit a reduced Fibonacci sequence?

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References

- [1] Alfred, U. (1964). Primes which are factors of all Fibonacci sequences. *Fibonacci Quart.* 2(1): 33–38.
- [2] Brison, O. J. (1992). Complete Fibonacci sequences in finite fields. *Fibonacci Quart.* 30(4): 295–304.
- [3] Brousseau, A. (1972). Table of indices with a Fibonacci relation. *Fibonacci Quart.* 10(2): 182–184. doi.org/10.1080/00150517.1972.12430957
- [4] Burr, S. A. (1971). On moduli for which the Fibonacci sequence contains a complete system of residues. *Fibonacci Quart.* 9(5): 497–504. doi.org/10.1080/00150517.1971.12430978
- [5] Campbell, C. M., Campbell, P. P., Doostie, H., Robertson, E. F. (2004). On the Fibonacci length of powers of dihedral groups. In: Howard, F. T., ed. *Applications of Fibonacci numbers*. 9: 69–85. doi.org/10.1007/978-0-306-48517-6_9

- [6] De Leon, M. J. (1977). Fibonacci Primitive Roots and the period of the Fibonacci numbers modulo p . *Fibonacci Quart.* 15(4): 353–55. doi.org/10.1080/00150517.1977.12430422
- [7] Halton, J. H. (1967). Some properties associated with square Fibonacci numbers. *Fibonacci Quart.* 5(4): 347–355. doi.org/10.1080/00150517.1967.12431289
- [8] E. T. Jacobson, Distribution of the Fibonacci numbers mod 2^k , *Fibonacci Quart.* **30** (1992) 211–215.
- [9] Knox, S. W. (1992). Fibonacci sequences in finite groups. *Fibonacci Quart.* 30(2): 116–120. doi.org/10.1080/00150517.1992.12429362
- [10] Lagarias, J. C. (1985). The set of primes dividing the Lucas numbers has density $2/3$. *Pacific J. Math.* 118(2): 449–461.
- [11] Lengyel, T. (1995). The order of the Fibonacci and Lucas numbers. *Fibonacci Quart.* 33(3): 234–239. doi.org/10.1080/00150517.1995.12429139
- [12] Li, H. C. (2000). Complete and reduced residue systems of second-order recurrences modulo p . *Fibonacci Quart.* 38(3): 272–281. doi.org/10.1080/00150517.2000.12428803
- [13] Li, H. C. (1999). Fibonacci Primitive Roots and Wall’s Question. *Fibonacci Quart.* 37(1): 77–84. doi.org/10.1080/00150517.1999.12428890
- [14] OEIS Foundation Inc. (2026). Entry A000057 in The On-Line Encyclopedia of Integer Sequences. oeis.org/A000057.
- [15] Renault, M. (1996). The Fibonacci sequence under various moduli (Master’s thesis). Wake Forest University, Winston-Salem, NC.
- [16] Shanks, D. (1972) Fibonacci primitive roots. *Fibonacci Quart.* 10(2): 162–168. doi.org/10.1080/00150517.1972.12430955
- [17] Tenenbaum, G. (2015). *Introduction to Analytic and Probabilistic Number Theory*. Graduate Studies in Mathematics. Vol. 163., 3rd ed. Providence, RI: Amer. Math. Soc.
- [18] Wall, D. D. (1960). Fibonacci series modulo m . *Amer. Math. Monthly.* 67(6): 525–532. doi.org/10.1080/00029890.1960.11989541